H. Vincent Kuo, Interim Department Head

Charles (Chuck) Stone

### Teaching Associate Professor

Emily Smith

### Teaching Assistant Professors

Victoria (Tori) Borish

Laith Haddad

Alysa (Ly) Malespina

### Research Professors

Wendy Adams

Charles (Chip) Durfee

### Research Associate Professor

K. Xerxes Steirer

### Research Assistant Professors

Kamil Ciesielski

Serena Eley

P. David Flammer

### Professors Emeriti

F. Edward Cecil

Reuben Collins

Thomas Furtak

Frank Kowalski

Mark Lusk

John Scales

P. Craig Taylor

John Trefny, President Emeritus

Lawrence Wiencke

Don Williamson

### Associate Professor Emeritus

David Wood

### Teaching Professor Emeritus

Todd Ruskell

# Interdisciplinary Programs

# Advanced Energy Systems

## Degrees Offered

- Master of Science in Advanced Energy Systems (non-thesis)
- Doctor of Philosophy in Advanced Energy Systems

## Program Description

The Advanced Energy Systems (AES) graduate engineering degree program is an interdisciplinary program designed in collaboration with researchers at the National Laboratory of the Rockies (NLR). The rigorous AES graduate program integrates the strengths of both institutions to offer MS and PhD students unique educational and research experiences.

The program is designed to prepare energy professionals and researchers to address a variety of complex problems associated with the energy transition. The breadth of potential research topics is considerable and spans topical areas including advanced generation technologies, grid modernization, energy security and resilience, critical minerals, building and renewable energy systems, electrochemical processes, technology integration, power markets, and power electronics to name a few. A common thread is the application of science and engineering fundamentals to develop effective solutions in any of the many dimensions needed to secure a sustainable energy future.

AES students are part of a cohort-based community of students, Mines faculty, and NLR technical staff that fosters professional development, cross-disciplinary thinking and systems understanding of the grand energy challenges. Educational emphases intermingle fundamental STEM coursework with techno-economic and environmental impact analysis to broaden student understanding of the entire energy supply and utilization value chain.

Graduates of the AES program are uniquely situated to enter the workforce in roles supporting advanced energy innovation and energy systems research and integration in government, academia, non-profit organizations, and the private sector.

## Program Description

The Advanced Energy Systems graduate engineering degree program offers the Master of Science and the Doctor of Philosophy of Advanced Energy Systems. The master’s program is a course-based, non-thesis program designed to prepare graduates for diverse careers in industry, government, and non-profit organizations or for additional graduate study at the PhD level.  The PhD degree program prepares graduates for careers in academia, industry, government, or non-profit leadership.

Prospective AES students apply to enter the Advanced Energy Systems program through Mines' Graduate Admissions. Final admissions decisions are made by AES after a holistic admissions review. MS and PhD curricula have overlap where new students in both programs complete three core courses as part of the degree program. Three core courses are required for all MS-NT students and all PhD students. Two additional core courses are required of all PhD students.

The following information provides detail on each of the degree options.

## Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

## Program Requirements

*Admitted Students:* The Advanced Energy Systems graduate admissions committee may require that any admitted student complete undergraduate remedial coursework to overcome technical deficiencies. Such coursework may not count toward the graduate degree. The AES admissions committee will decide whether to recommend regular or provisional admission. In accordance with Mines institutional policy, all AES students are required to maintain a minimum cumulative GPA of 3.0.

*Transfer Courses*: Graduate-level courses taken at other universities for which a grade equivalent to a B or better was earned may be considered for transfer credits for the AES MS and PhD degree program. Approval must be granted after all appropriate documentation has been submitted to the AES program director who will conduct a review of the transfer petition. Review of transfer credits is based on the relevance of the coursework to AES and the recency of completion for any transfer credits requested. Some exceptions may apply, however, coursework completed more than five years from AES enrollment will not be considered. Transfer credits may not have been used as credit toward a bachelor's degree. For the MS degree, no more than 9 credits may transfer. For the PhD degree, up to 24 credits may be transferred. Students who enter the PhD program with a thesis-based master's degree from an accredited institution may request to transfer up to 36 hours in recognition of the coursework and research completed for that degree. All transfer credits are to be submitted in the first semester of enrollment for program consideration and review for recency and AES relevance.

## Master of Science – Non-Thesis Degree Requirements

The AES Master of Science, Non-Thesis (MS-NT) is a stand-alone graduate engineering degree wherein students are self-supported or supported by industry or other outside sources. The AES MS-NT degree is course-based and targeted for students interested in professional careers in industry, government, or non-governmental positions. Students enrolled in the MS program do not participate in the three 600 level PhD core. The MS-NT degree requires 30 credit hours of coursework as stipulated below. Approved elective credit hours must be taken at the graduate level at Mines and are reviewed to ensure that they form a coherent focus for an in-depth energy science and engineering study.

*Advisor:* All AES MS students will have an advisor to guide and monitor their academic plan and progress toward graduation.

## MS Non-Thesis Degree

| | | |
|---|---|---|
| ENGY501 | PHYSICS OF ENERGY RESOURCES & CONVERSION | 3.0 |
| ENGY502 | ENERGY FOR TRANSPORTATION | 3.0 |
| ENGY503 | ENERGY & POWER SYSTEMS INTEGRATION | 3.0 |
| Economics | Student must select one graduate-level Economics course from the approved program list. | 3.0 |
| Technical Electives | Technical Electives (STEM Courses approved by Advisor and Program Director) | 12-18 |
| Economics or Policy | Student may select up to two (2) courses from the approved program list. | 0-6 |
| Total | | 30.0 |

## Advanced Energy Systems Doctor of Philosophy Degree Requirements

The PhD degree in Advanced Energy Systems requires 72 total credits. A minimum of 36 credits of coursework and 36 credits of research credits must be completed. A minimum of 15 of the 36 credits of required coursework must be taken at Colorado School of Mines as three core courses plus the three 600-level PhD courses.

The AES PhD program is designed as a full-time, four-year program of study for students having completed a minimum of a bachelor's degree in Science or Engineering. In accordance with all other graduate programs at Mines, a minimum of two semesters of full-time enrollment is required and follows the program as detailed below.

## PhD Degree Requirements

| | | |
|---|---|---|
| ENGY501 | PHYSICS OF ENERGY RESOURCES & CONVERSION | 3.0 |
| ENGY502 | ENERGY FOR TRANSPORTATION | 3.0 |
| ENGY503 | ENERGY & POWER SYSTEMS INTEGRATION | 3.0 |
| ENGY691 | INTRODUCTION TO RESEARCH METHODS IN THE ENERGY SCIENCES | 3.0 |
| ENGY692 | PROJECT FOCUSED RESEARCH IN ENERGY SCIENCE & TECHNOLOGY | 3.0 |
| Economics | Student must select one graduate-level Economics course from the approved program list | 3.0 |
| Technical Electives | Technical Electives (STEM courses approved by Advisor and Program Director) | 15-18 |
| Economics or Policy Elective | Student may select one graduate-level Economics or Policy course from the approved program list | 0-3 |
| ENGY693 | AES GRADUATE STUDENT SEMINAR (AES PhD Seminar: Students are required to register every fall and spring. Students may use up to 3 credits towards the degree.) | 0-3 |
| ENGY707 | GRADUATE THESIS | 36.0 |
| Total | | 72.0 |

*Timeline and Milestones:* PhD students must make adequate progress and reach appropriate milestones toward their degree by working with their faculty advisor and thesis committee. The AES Program has adopted a standard Mines PhD timeline that outlines milestones that students are expected to reach on a semester-by-semester basis. Each milestone is listed here with detail.

| PhD MILESTONE | EXPECTED TIMELINE |
|---|---|
| Select a permanent advisor | By Second semester |

| | |
|---|---|
| Complete the PhD qualifying examination | In June of first year of enrollment |
| Establish a dissertation committee | By Fourth semester |
| Complete all core curriculum course requirements | Fourth semester |
| Present research proposal; Submit Degree Audit and Admission to Candidacy forms | Fifth semester or earlier |
| Present a preliminary defense | 9-12 months before dissertation defense |
| Present a dissertation defense | End of Year Four |

*Funding:* Students admitted to the AES PhD program are fully funded. Typically, first-year funding is provided by the Program. Subsequent funding is provided for the subsequent three years by the research group to which the PhD student is aligned.

*Advisor : Preliminary and permanent:* AES students are assigned a first-year advisor to guide and monitor their progress toward the qualifying exam. First-year PhD advisors are assigned based on the student's emergent research interests stated at the time of application. By the end of the first year, students select a permanent academic advisor closely aligned with their academic and research interests, who will serve to guide, monitor, and evaluate their progress toward the final dissertation defense. A unique aspect of the AES program is the NREL mentors who are engaged to support the student's research efforts. While the lead advisor is the Mines academic graduate faculty, the NREL research provides additional guidance for the student's research agenda.

*Qualifying Exam:* All AES PhD students are required to pass a qualifying exam given at the end of the second semester of enrollment. PhD students must have earned a minimum grade point average of 3.3 and have completed the first-year required core courses to sit for the qualifying exam. The AES program director oversees the process and ensures that the exam is administered fairly.

In accordance with other PhD programs at Mines, the purpose of the qualifying exam is to assess attributes expected of a successful PhD student, including:

- Ability to review, synthesize and apply fundamental concepts.
- Creative and technical ability to propose research solutions to solve open-ended and challenging problems.
- Technical Communication Skills.

The AES qualifying exam requires students to develop a hypothesis-driven research proposal on the student's topic of interest in consultation with the academic advisor and research mentor. Students submit their research proposal to a panel of reviewers to include the AES program directors and one invited subject matter expert. The student presents the research proposal and, similar to the thesis-proposal process, answers a series of questions from the panel.

Exam results of pass, conditional pass, or fail are provided to the students immediately following the oral defense portion of the exam. Written feedback is provided within three days of the exam. Students who fail the exam may be offered one additional opportunity to pass the exam before the start of the third semester. Students who fail on the second attempt may be offered an opportunity to complete the MS-NT as the terminal degree and may be provided one immediate final semester of support to do so.

*Thesis Committee:* Upon successful completion of the qualifying exam and in consultation with the Mines academic advisor, each PhD student will assemble a five-person thesis committee that includes: the student's Mines academic advisor and NREL mentor (as the student's co-advisor); two additional Mines faculty who have a science and/or engineering background; and a fifth member of the committee who serves as the chair and is not in the home department of the advisor. Additional committee members with specific expertise may be allowed. If there is no NREL mentor, the committee composition may constitute four members, minus the co-advisor. The AES program director must review and approve all committee requests prior to submission to the Office of Graduate Studies.

*Research Proposal:* After passing the qualifying exam, the PhD student will prepare a written research proposal for the dissertation and present it formally to the dissertation committee, which will have been selected by the student in consultation with the Mines academic advisor and approved by the program director.

A written research proposal document will be provided to the committee in advance of the presentation with the expectation of achieving the following:

- Demonstrate a knowledge of background and motivation of the research problem being undertaken as embodied by a review of the relevant literature.
- Enumerate specific aims and/or hypotheses.
- Identify preliminary techniques, materials, and specific measurements for the proposed research project.
- Explain clearly the scientific merit (value added) of the proposed work.
- Provide a general idea of the timeline for the research program.
- Specify potential publications and presentations that may arise from the work.

The student and the advisor must convene a meeting of the full dissertation committee in which the student gives an oral summary of their written proposal in a 30- to 45-minute presentation. This research proposal gives the committee an early chance to discuss the work and to help the student more clearly define the work and identify the salient aspects. The research proposal presentation should ideally occur before the beginning of year 3 of the program. The research proposal must be completed before admission to candidacy.

*Degree Audit and Admission to Candidacy:* PhD students must complete the Degree Audit form by the posted deadlines and the Admission to Candidacy form by the first day of classes of the semester in which they want to be considered eligible for reduced registration.

Additionally, full-time PhD students must complete the following requirements within the first two calendar years after enrolling in the AES PhD program:

- Have a thesis committee appointment form on file in the Graduate Office.

- Complete all prerequisite and core curriculum course requirements.
- Demonstrate adequate preparation for and satisfactory ability to conduct doctoral research.
- Be admitted into full candidacy for the degree.

*Preliminary Defense:* Prior to the final dissertation defense, the PhD student will make an oral presentation to the student's committee to summarize research accomplishments and remaining goals and work plan. This meeting serves as a final check to assess if the student's progress is on schedule for graduation. This meeting should present a preliminary document that will likely evolve and expand into the dissertation. The preliminary document should include basic literature review, methodologies used, results to date, and an estimated timeline for remaining work. The student must give no more than a 45-minute presentation that summarizes the work already accomplished, including their relevant publication(s) and a proposed plan of the work needed to culminate in a formal defense and graduation. The committee will provide feedback and, as necessary, revisions to the proposed work plan such that its completion should lead to a successful dissertation defense and publication record in a realistic time frame. The time period between the research proposal and the preliminary defense can span a few years, but the preliminary defense should take place 12 months and no less than six months prior to the date of dissertation defense.

Publications and Presentations: The required and recommended journal publications for PhD students prior to graduation are listed below. Students wanting to defend before meeting these requirements must submit a one-page petition with a reasonable explanation to the AES program director.

*Journal Publications: Required:* Minimum of two first-author paper accepted or published (DOI is required) in a peer-reviewed journal (recognized as high quality in the research field), before Dissertation Defense. *Recommended:* Three or more first-author papers accepted or published in peer-reviewed journals. More than three first author journal publications are recommended for students interested in academic positions.

*Presentations — Required:* Minimum of one research presentation (poster or podium) at an external technical conference before the dissertation defense. Minimum of three presentations in the AES graduate seminar or equivalent (such as campus-wide graduate student research conference, research sponsor meetings, or additional conference presentations) during PhD program. *Recommended:* Two or more conference presentations (poster or podium) before the dissertation defense in which the student is the first author on these presentations. Numerous conference presentations are strongly encouraged to establish a reputation amongst researchers in a field for students interested in academic positions.

*Thesis Defense:* At the conclusion of their PhD program, students are required to make a formal presentation and defense of their thesis research. A student must "pass" this defense to earn a PhD degree. The dissertation document should be submitted to the dissertation committee at least 10 days prior to the defense. The committee will perform a post-presentation review of the dissertation, technical contributions, and publications with the student. The committee may request revisions to the dissertation and additional work that requires subsequent review by the advisor and or the committee.

*Unsatisfactory Progress*: To ensure that a student receives proper feedback if progress toward the preliminary defense or the dissertation defense is not satisfactory, the advisor must provide the student and the committee a brief, written progress evaluation. If the student's progress is unsatisfactory such that the advisor gives them a PRU grade for research credits, the student will go on academic probation as outlined in the graduate bulletin.

## Courses

**ENGY501. PHYSICS OF ENERGY RESOURCES + CONVERSION. 3.0 Semester Hrs.**

This course will provide successful students a quantitative understanding of how fossil, renewable and nuclear energy resources are harnessed to electric power. A foundational underpinning will be the thermodynamics of energy conversion, using fundamental principles and language bridging physics, chemistry and engineering. Examples will be taken from both established and emerging technologies spanning solar, nuclear, wind fossil fuel and bioenergy conversion. Students will also learn how to analyze electricity generation, transmission, and grid-scale storage systems with a focus on the U.S. as a framework for analyzing other developing markets.

**Course Learning Outcomes**

- With respect to all energy resources: 1. Articulate in writing, major quantitative trends and forces behind the evolution of fossil, nuclear, and renewable energy resource utilization of the U.S.; 2. Demonstrate knowledge of how permitting, regulations, markets, and environmental impact drive investment for utility-scale power generation; 3. Compare quantitatively the levelized cost of electricity, major environmental impacts, and water usage for electricity generation from fossil, renewable, and nuclear resources;
- With respect to solar resources: 4. Quantitatively assess solar resources in the U.S. and the challenges (economic and grid-related) to increase solar penetration on the electric grid; 5. Quantify advances in solar photovoltaic technology in terms of cost and efficiency and estimate trends for levelized costs of electricity for photovoltaics over the next 10 years; 6. Evaluate levelized cost of electricity for concentrating solar power with thermal energy storage for utility scale applications;
- With respect to wind, geothermal, and hydroelectric resources 7. Quantitatively assess wind, geothermal, and hydropower resources in the U.S. and opportunities for increased penetration in the electric utility grid; 8. Perform basic analysis of levelized cost of electricity as a function of wind conditions, turbine size, and turbine location based on weather conditions;
- With respect to biomass resources 9. Compare different biomass resources for fuels and power plants and their economic viability relative to other fuel and power resources;
- With respect to fossil fuel resources: 10. Demonstrate quantitative knowledge of coal and natural gas extraction from an unconventional reservoir and evaluate costs of extraction and delivery of fossil fuel resources; 11. Analyze basic thermodynamic performance and operation of present-day Rankine, Brayton, and combined cycle power plants running on fossil fuel inputs including coal;
- With respect to nuclear resources: 12. Demonstrate basic knowledge of differences in the major nuclear power plant technologies in terms of conversion efficiencies and waste streams;
- With respect to the current utility-scale electric grid: 13. Evaluate the cost and performance of power generation sources and grid-scale storage technology and assess the impact on energy markets and renewable energy penetration; 14. Assess the impact of storage on transition of the US power system and future research

and development needs; 15. Perform real-time electricity pricing calculation for a selected electricity market in U.S.;

- With respect to advanced energy grids, distributed power, and energy storage: 16. Articulate advances in distributed power and advanced controls and their impact upon grid operation and utility markets in a smart grid framework for the U.S.; 17. Identify the challenges and risks arises due to cyber-security in a smart grid and articulate awareness of the essential tools used to secure grid infrastructure; 18. Evaluate energy storage technologies and compare their economic feasibility, round-trip efficiency, and potential capacity for distributed power applications.

**ENGY502. ENERGY FOR TRANSPORTATION. 3.0 Semester Hrs.**
(I) This course focuses on multiple aspects of current and proposed transportation technologies to analyze the challenges and opportunities of moving toward more sustainable transportation infrastructure. This course is designed to train students to develop analytical skills and to use computational tools for evaluating performance and environmental impacts of various vehicle and fueling technologies. Successful students will develop a basis for assessing energy resource requirements and environmental concerns within the context of technical performance, policy frameworks, and social perspectives. The course will include the following topics: travel demand and travel modes; transportation technologies; fossil-fuel and electric power plants and associated fuels; emissions (CO2 and pollutants) formation and impacts on air quality, climate, and human health; national/international transportation policy; and transportation planning. 3 hours lecture; 3 semester hours.
**Course Learning Outcomes**

- With respect to the oil and gas distribution systems: 1. Perform assessment on the oil and natural gas distribution network in terms of sensitivity of fuel prices to key economic, environmental and political factors; 2. Evaluate the midstream and downstream production of U.S. liquid fuels; 3. Articulate the risks associate with fossil fuel production and utilization and the associated environmental impacts; 4. Assess the impact of vehicle electrification on oil and gas markets;
- With respect to aerospace and shipping transportation and fuels: 5. Assess the efficiency of all types of vehicle transportation technology and major historical trends with respect to fuel demands for various transportation sectors; 6. Identify fuel(s) needed for aircraft propulsion and assess trends for efficiency and reduced emissions from large aircraft; 7. Assess fuel utilization for large-scale shipping of goods with ships, trains, and large trucks and evaluate potential paradigm shifts in heavy transportation; 8. Perform well-to-wheel analysis on at least two large-vehicle technology platforms and assess the feasibility of alternative fuel sources and/or energy carriers;
- With respect to automotive transportation, fuels, and alternative energy carriers: 9. Assess internal combustion engine technology performance and emissions from a historical perspective and with respect to current-day trends; 10. Evaluate electric vehicle performance including battery and charging technologies including with respect to costs and energy demands for selected transportation sectors; 11. Articulate opportunities and challenges of a hydrogen- or biomass/waste gasification to support fueling infrastructure for a region of the U.S.; 12. Perform well-to-wheel analysis on at least two automotive technology platforms and assess their impact on energy usage and emissions; 13. Conduct technoeconomic analysis of hydrogen as an energy carrier for the transportation sector and the potential for fuel cells in vehicles; 14. Identify the challenges and risks arises due to cyber-security in an electrified transportation system, including autonomous systems.

**ENGY503. ENERGY & POWER SYSTEMS INTEGRATION. 3.0 Semester Hrs.**
This course will provide students with basic skills to analyze the operation and evolution of the electric grid and electricity utilization with a particular emphasis on trends toward increased renewable energy penetration. The course will develop students' analytical skills to evaluate how electricity generation, transmission, distribution and storage are managed and controlled. Successful students will gain a basic understanding of electromechanical machines for power conversion and AC power distribution as well as renewable energy sources and battery systems with DC storage. The course will introduce students to how efficient energy utilization and demand response management impact the electric grid performance and electricity markets. An emphasis on managing energy loads in buildings, the commercial sector, and energy-intensive manufacturing will expose students to system-level modeling tools that can assess how to manage power demands with transient power generation and market forces. The course will also address the integrated nature of energy systems with an emphases on connections to water demands and on risks arising due to cybersecurity and resiliency threats facing the electric grid.
**Course Learning Outcomes**

- With respect to the current broad trends in energy utilization in buildings : 1. Assess quantitative trends and major forces behind the past and future evolution with respect to energy utilization in residential buildings, commercial sector, and major industries; 2. Analyze cooling and heating loads of residential buildings and commercial buildings as a function of HVAC technology and geographical location; 3. Evaluate historical trends and new technologies in reducing and increasing electric, heating, and cooling demands of residential and commercial buildings;
- With respect to dynamic energy utilization for buildings: 4. Quantify the impacts of broad implementation of energy storage such as batteries and/or thermal energy storage on the impact of electric grid requirements; 5. Quantify how dynamic demand response or load management with energy storage can impact energy requirements for buildings/communities; 6. Describe the role of LEED Certification in driving building energy efficiency;
- With respect to energy utilization for industrial processes: 7. Analyze two energy intensive manufacturing processes (chemical or materials) and write review of technologies that impact their energy needs, requirements, and costs over the next two decades; 8. Develop understanding of how changing energy and utility supplies can impact process efficiency and environmental impact on energy intensive manufacturing processes; 9. Articulate the impact of PURPA (Public Utility Regulatory Policies Act), and Energy Policy Acts of 1992, and 2005.
- With respect to advanced energy grids and distributed power: 10. Articulate advances in distributed power and advanced controls and their impact upon grid operation and utility markets in a smart grid framework for the U.S.; 11. Evaluate for one energy-intensive industrial process the energy, economic benefits for combined heat and power or combined cooling, heat, and power; 12. Identify the challenges and risks arises due to cyber-security in a smart grid and articulate awareness of the essential tools used to secure grid infrastructure.

**ENGY550. FUNDAMENTALS OF SOLAR ENERGY ENGINEERING. 3.0 Semester Hrs.**
This course will go over the solar resource and components of solar irradiance; principles of photovoltaic devices; wafer-based and thin film photovoltaic modules; photovoltaic system design; photovoltaic electrical

energy production and cost analysis of photovoltaic systems relative to fossil fuel alternatives; integration into conventional energy systems; introduction to concentrated photovoltaic systems, solar water heating, solar ventilation air preheating, and passive solar building design. Case studies include net zero residential and commercial buildings and utility-scale solar plants. Prerequisites: PHGN200 and MATH225.

**Course Learning Outcomes**

- Derive relations based on key concepts of physics (quantum and spectral concepts; boundary layer theory; energy balance; heat exchanger effectiveness) to design and analysis of solar energy systems.
- Derive relations based on key concepts of physics (quantum and spectral concepts; boundary layer theory; energy balance; heat exchanger effectiveness) to design and analysis of solar energy systems.
- Understand the operating principle of key solar energy technologies.
- Understand the function of each component of a solar energy system and how components are assembled into systems.
- Practice handbook and computer methods to estimate solar system energy delivery.
- Understand the life-cycle economics and financing strategies for solar energy systems and practice cost-estimating.
- Understand the financing and contracting process by which solar energy systems are delivered.
- Instill awareness of the effects of a solar energy system on the larger utility system and on the environment.

**ENGY598. SPECIAL TOPICS. 0-6 Semester Hr.**

**ENGY598. SPECIAL TOPICS. 0-6 Semester Hr.**

**ENGY598. SPECIAL TOPICS. 0-6 Semester Hr.**

**ENGY599. INDEPENDENT STUDY. 0.5-6 Semester Hr.**
Individual research or special problem projects supervised by a faculty member, also, when a student and instructor agree on a subject matter, content, and credit hours. Prerequisite: “Independent Study” form must be completed and submitted to the Registrar. Variable credit: 0.5 to 6 credit hours. Repeatable for credit under different topics/experience and maximums vary by department. Contact the Department for credit limits toward the degree.

**ENGY691. INTRODUCTION TO RESEARCH METHODS IN THE ENERGY SCIENCES. 3.0 Semester Hrs.**
This course introduces graduate students enrolled in the Advanced Energy Systems Program to research opportunities, culture, and expectations in energy science and technology with a particular emphasis on systems and/or policy analysis. Students will work within directorates at NREL with an emphasis on systems modeling, analysis, and/or integration. This class will engage students in a semester-long research project in energy system analysis and prepare students for best practices with respect to research project and data management, literature reading, report writing, and presentation.

**Course Learning Outcomes**

- With respect to research skills and practices: 1. Learn how to perform a detailed literature review and summary in support of a research project; 2. Develop strong habits for documenting research progress and reporting to other researchers and management on regular basis; 3. Exhibit working knowledge of good practices in data management during research; 4. Demonstrate competency in presenting research in a professional, technical presentation;
- With respect to energy systems analysis: 5. Perform a publication-quality techno-economic or systems analysis to assess energy-relevant technology, innovation, or policy for informing decision-making; 6. Leverage existing models and develop new models to evaluate advanced energy technologies, systems, and services.

**ENGY692. PROJECT FOCUSED RESEARCH IN ENERGY SCIENCE & TECHNOLOGY. 3.0 Semester Hrs.**
(I) This course prepares graduate students enrolled in the Advanced Energy Systems Program in research practices, culture, and expectations in energy science and technology with a particular emphasis on science and engineering related to energy materials, processes, and/or systems. Students will work within directorates at NREL with an emphasis on science and/or technology. This class will engage students in a semester-long research project in energy science and/or technology. Students will also learn and practice journal publication and research poster best practices, research career path planning, and proposal funding strategies. 1 hour lecture; 6 hours lab; 3 semester hours.

**Course Learning Outcomes**

- With respect to journal article and poster preparation: 1. Develop strong skills in literature review and paper preparation; 2. Demonstrate skills with respect to preparing a publication outline and figure list; 3. Write a report that has the structure and quality of a journal publication; 4. Prepare a poster summarizing research motivation and results from semester project;
- With respect to career development and supporting skills: 5. Articulate career paths related to advanced energy systems and identifying skills for preparation 6. Prepare and critique proposal project summaries for energy-related science and technology research; 7. Demonstrate ability to identify funding opportunities and promote research;
- With respect to energy science, technology, and innovation research: 8. Perform a research study on an advance energy science and technology development in line with career and research interest and NREL programs; 9. Demonstrate working knowledge of best laboratory practices with respect to safety, notebook recording, and uncertainty. 10. Be able to motivate the topic of research through techno-economic analysis 11. Be able to define a research hypothesis and develop a parametric test procedure to evaluate it

**ENGY693. AES GRADUATE STUDENT SEMINAR. 0.5 Semester Hrs.**
The Advanced Energy Systems Graduate Student Seminar is a series of presentations provided by graduate students to fellow graduate students, faculty, mentors, and guests. All Ph.D. students are expected to register for this course. The seminar course provides students, faculty, and mentors working in the AES Graduate Program an opportunity to hear updates on current research within the various cohorts and provides a chance for students to get constructive feedback on their presentation. In addition, the course will provide a venue for discussions on various topics related to methods for succeeding in research careers in academia, national labs, and industry, and topics of the day. The course format will be to have two graduate-student presentations with critical feedback, followed by a discussion session on various professional development topics.

**ENGY707. GRADUATE THESIS. 1-15 Semester Hr.**

## Professor and Program Director

Robert Braun, University Distinguished Professor, Mechanical Engineering

## AES Steering Committee

Robert Braun (ME), Director and Chair

Morgan Bazilian (EB)

Tzahi Cath (CEE)

Steven DeCaluwe (ME)

Greg Jackson (ME)

Michael Kaufman (MME)

Monica Kosanovich (AES)

Stephanie Kwon (ChemE)

Salman Mohagheghi (EE)

Alexandra Newman (OR)

Ryan O'Hayre (MME)

Matthew Posewitz (Chem)

Vladan Stevanovic (MME)

Paulo Tabares Velasco (ME)

Jeramy Zimmerman (PHYS)

## NRL AES Steering Committee

Kate Anderson, AES Program Co-Director

## Interdisciplinary Talents

In addition to the steering committee members, the AES program engages the interdisciplinary expertise of colleagues from various departments at Mines and NREL who serve as research mentors, co-instructors, academic advisors and committee members.

# Additive Manufacturing

## Program Description

The Additive Manufacturing program provides graduates and professional students with the practical, interdisciplinary skills to apply cutting-edge additive manufacturing techniques to a wide range of industries, including aerospace, biomedical, defense, and energy, among others.

This program highlights the core foundations of additive manufacturing: process, design, and materials. Students can choose to from a wide range of electives across a variety of categories including data science and machine learning, operational efficiency of manufacturing operations, and entrepreneurship and technology management. The program focuses on additive manufacturing of structural materials and provides users with the skills they need to apply additive manufacturing in modern manufacturing environments.

## Primary Contact

Craig Brice
Director, Additive Manufacturing
https://additive.mines.edu/

**Master of Science in Additive Manufacturing (Non-Thesis) (30 credits)**

The Master of Science in Additive Manufacturing (non-thesis) provides students the opportunity to explore a wide range of manufacturing technologies and methodologies necessary to fabricate engineered products in current and emerging markets.

The core courses in the Additive Manufacturing program give students a foundation in additive manufacturing along with options to explore topics across various fabrication and manufacturing efficiency areas. Students enrolled in the Master of Science in Additive Manufacturing (non-thesis) program will complete 12 credits of core courses listed below along with 21 credits of elective courses from the Additive Manufacturing list. The elective list is broken into focus areas to aid students interested in a specific area of additive manufacturing. Students are not required to select a focus area but can choose electives across the entire list.

| | | |
|---|---|---|
| AMFG501 | ADDITIVE MANUFACTURING PROCESSES | 3.0 |
| AMFG521 | DESIGN FOR ADDITIVE MANUFACTURING | 3.0 |
| AMFG531 | MATERIALS FOR ADDITIVE MANUFACTURING | 3.0 |
| AMFG541 | ADDITIVE MANUFACTURING PROJECT | 3.0 |
| REQUIRED ELECTIVES | Select two electives from the required list below | 6.0 |
| ADDITIONAL ELECTIVES | Select electives from the Additive Manufacturing list below Up to 6 hours may be replaced with project-based independent study | 12.0 |
| **Total Semester Hrs** | | **30.0** |

## Mines' Combined Undergraduate/Graduate Degree Program

Students enrolled in Mines' combined undergraduate/graduate program may double count up to 6 credits of graduate coursework to fulfill requirements of both their undergraduate and graduate degree programs. These courses must have been passed with B- or better, not be substitutes for required coursework, and meet all other university, department, and program requirements for graduate credit.

Students are advised to consult with their undergraduate and graduate advisors for appropriate courses to double count upon admission to the combined program.

## Additive Manufacturing Electives

Note that, while the listed electives are grouped into focus areas, students are not required to select a specific focus area for their degree.

| | | |
|---|---|---|
| REQUIRED ELECTIVES (choose two) | | |
| AMFG511 | DATA DRIVEN ADVANCED MANUFACTURING | 3.0 |
| AMFG522 | LEAN MANUFACTURING | 3.0 |
| AMFG523 | DESIGN AND ANALYSIS OF EXPERIMENTS | 3.0 |
| MEGN579 | OPTIMIZATION MODELS IN MANUFACTURING | 3.0 |
| ADDITIONAL ELECTIVES | | |
| **Materials Manufacturing Focus Area** | | |
| MEGN511 | FATIGUE AND FRACTURE | 3.0 |
| MEGN515 | COMPUTATIONAL MECHANICS | 3.0 |
| MTGN531 | THERMODYNAMICS OF METALLURGICAL AND MATERIALS PROCESSING | 3.0 |
| MTGN536 | OPTIMIZATION AND CONTROL OF METALLURGICAL SYSTEMS | 3.0 |